

EXPLORATORY DATA ANALYSIS

E-Commerce Orders Dataset

Project 2 | DecodeLabs Data Analytics Internship | Batch 2026

Prepared by: Fatima

BS Biostatistics | Punjab University Lahore
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1. Problem Statement

This report presents a comprehensive Exploratory Data Analysis (EDA) of an e-commerce orders dataset provided by DecodeLabs. The core business question is:

"What patterns, trends, and anomalies exist in our e-commerce orders, and what do they mean for business decision-making?"

By analyzing 1,000 orders across multiple dimensions — products, payment methods, referral sources, and order statuses — this analysis aims to uncover actionable insights that can guide marketing strategy, inventory planning, and customer experience improvements.

2. Methodology

The following EDA framework was applied to the dataset:

- Data Inspection: Reviewed all 14 columns and 1,000 rows for structure and completeness
- Descriptive Statistics: Calculated mean, median, min, max, and count for numeric variables
- Outlier Detection: Applied the IQR (Interquartile Range) method to identify anomalous order values
- Trend Analysis: Used PivotTables to break down revenue and order counts by Product, Order Status, Referral Source, and Payment Method
- Insight Synthesis: Applied the 'So What?' test to translate statistical findings into business recommendations

Tools used: Microsoft Excel (PivotTables, QUARTILE, COUNTIF, AVERAGE, MEDIAN functions)

3. Descriptive Statistics

The table below summarizes the key statistical measures for the dataset:

Metric	Value
Total Orders	1,000
Total Revenue	PKR 1,051,995
Average Order Value (Mean)	PKR 1,051.99
Median Order Value	PKR 797.49
Maximum Order Value	PKR 3,456.40
Minimum Order Value	PKR 14.06

Average Quantity per Order	~3 items
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Key Statistical Insight

The mean (PKR 1,051.99) is significantly higher than the median (PKR 797.49). This gap of PKR 254 indicates that the data is right-skewed — a small number of very high-value orders are pulling the average upward. This means the median is a more reliable measure of a "typical" order for this business.

4. Outlier Analysis

The IQR (Interquartile Range) method was used to detect outliers in the TotalPrice column:

Measure	Value
Q1 (25th Percentile)	Calculated via Excel QUARTILE()
Q3 (75th Percentile)	Calculated via Excel QUARTILE()
IQR (Q3 - Q1)	Calculated
Lower Bound (Q1 - 1.5 x IQR)	No outliers detected (Count = 0)
Upper Bound (Q3 + 1.5 x IQR)	5 orders flagged as outliers

Outlier Insight

5 high-value orders were detected above the upper bound, while no unusually low orders were found. These 5 outliers likely represent bulk purchases or premium product orders. They should be investigated further — they could represent VIP customers or corporate buyers worth targeting with special offers.

5. Product Analysis

Revenue and order count breakdown by product category:

Product	Total Revenue (PKR)	Observation
Chair	195,620.11	Highest Revenue
Printer	195,612.61	Almost equal to Chair
Monitor	~190,000+	Strong performer
Tablet	~180,000+	High volume
Laptop	~170,000+	Mid range
Desk	~150,000+	Steady
Phone	Lowest	Lowest Revenue

Product Insight

- Chair leads with PKR 195,620 in revenue — but only by PKR 7.50 over Printer (PKR 195,612). This near-tie means both products are equally critical to the business.
- Phone generates the lowest revenue despite being a common product — this may indicate low unit prices or low quantities ordered.
- Business implication: Inventory and marketing efforts should not focus on just one product. Chair and Printer are co-leaders and need equal attention.

6. Order Status Analysis

Distribution of order statuses across 1,000 orders:

Order Status	Count
Delivered	231
Cancelled	250
Returned	247
Shipped	234
Pending	237

Order Status Insight

- Cancelled orders (250) and Returned orders (247) together account for nearly 50% of all orders — this is a critical red flag.
- Almost 1 in 2 orders either never completes or comes back. This directly impacts revenue and increases operational costs.
- Business implication: Urgent investigation needed into WHY customers are cancelling and returning. Is it product quality? Delivery delays? Wrong expectations set by product listings?

7. Referral Source Analysis

Analysis of which marketing channels drive the most traffic and revenue:

Referral Source	Order Count	Revenue
Instagram	259 (Top)	Highest
Facebook	~210	High
Google	~190	High
Email	~180	Moderate
Referral (Word of Mouth)	~161	Moderate

Referral Source Insight

- Instagram is the top-performing channel — both in number of orders (259) and total revenue. This confirms Instagram as the primary customer acquisition platform.
- Business implication: Marketing budget should prioritize Instagram. Sponsored posts, influencer collaborations, and Instagram Shopping features should be explored to further capitalize on this channel.

8. Payment Method Analysis

Breakdown of preferred payment methods used by customers:

Payment Method	Observation
Credit Card	Most preferred — Top payment method
Debit Card	Second most used
Online (Net Banking)	Moderate usage
Cash on Delivery	Moderate usage
Gift Card	Least preferred

Payment Method Insight

- Credit Card dominates as the preferred payment method, suggesting customers trust the platform with card transactions.
- Business implication: Ensure seamless credit card payment experience. Consider offering cashback or reward points for credit card users to further incentivize this behavior.

9. Key Findings Summary

The following are the strongest signals identified through this EDA:

#	Finding	Business Diagnosis
1	Data is right-skewed (Mean > Median)	Standard average is misleading; use median for realistic business targets
2	5 high-value outlier orders detected	Potential VIP/corporate customers worth targeted outreach
3	Chair & Printer almost equal in revenue	Both products are equally critical; neither should be deprioritized
4	~50% orders Cancelled or Returned	Major revenue leakage; requires immediate operational investigation
5	Instagram is top acquisition channel	Maximize Instagram marketing ROI with dedicated budget allocation
6	Credit Card is most preferred payment	Optimize and reward credit card checkout experience

10. Recommendations

Based on the EDA findings, the following actionable steps are recommended:

Immediate Actions (0-30 days)

- Investigate cancellation & return reasons — conduct customer surveys or analyze cancellation notes to identify root causes of the ~50% incomplete order rate.
- Flag and personally reach out to the 5 outlier (high-value) customers — they may be corporate buyers who can be converted into repeat bulk purchasers.

Short-Term Strategy (1-3 months)

- Double down on Instagram marketing — allocate dedicated budget for Instagram ads, influencer partnerships, and Instagram Shopping integration.
- Introduce Credit Card rewards — loyalty points or cashback for credit card payments to encourage repeat purchases.

Long-Term Strategy (3-6 months)

- Balance product promotion — since Chair and Printer are nearly tied in revenue, design bundled offers (e.g., 'Office Setup Bundle: Chair + Printer') to boost both simultaneously.
- Use median (PKR 797) as baseline for pricing strategy — most customers spend around this amount, so promotions targeting the PKR 700-900 range will reach the widest audience.
- Monitor Phone category — investigate why Phone generates lowest revenue and consider whether pricing, product quality, or marketing is the issue.

11. Conclusion

This Exploratory Data Analysis of 1,000 e-commerce orders has successfully uncovered six key business signals from raw data. The most critical finding is that nearly 50% of orders are either cancelled or returned — a major operational challenge that must be addressed to protect revenue. At the same time, Instagram has emerged as the dominant growth channel, and Chair/Printer as the top revenue-generating products.

As the DecodeLabs training material emphasized: EDA is not about reporting numbers — it is about translating noise into signal and data into decisions. This analysis has done exactly that, providing a clear roadmap for the next steps in this business.
